

Current trends in computer vision

Computer vision on edge. To successfully deploy computer vision on edge, neural network design must be integrated into embedded platforms. Then, there are also difficult design decisions to be made around power consumption, cost, accuracy, and flexibility. Overall, for bandwidth, speed, and security reasons, edge computing is critical to computer vision processes. Still, hackers could take advantage of emerging security flaws in artificial intelligence. Designers must take this into consideration too.

However, there are some challenges when it comes to vision AI on edge, such as involving all edge endpoints can lead to complex device architectures. This makes device management and edge computing implementation harder. Edge computing differs from cloud architecture as it is spread out rather than centered around a single cloud. For large enterprises, it may be assumed that such an infrastructure will have higher maintenance costs than one centered upon the cloud.

Embedded vision and IoT embedded vision. Embedded vision is when we combine cameras with the processing unit, like smart cameras mentioned before. Many companies, like Lattice Semiconductor and Blaser AG, provide embedded vision solutions. For example, Basler's



Blaser's AI Vision Kit (Credit: Blaser AG)

AI Vision Solution Kit comes with cloud connectivity and contains a dart MIPI camera module, an NVIDIA Jetson Nano development board, lens, and cable. It allows developers to test AI based IoT applications on an optimised vision system and to access cloud services.

But we are about to enter the next stage of the Internet of Things. So far, the focus was on connecting devices, accumulating data, and creating big data platforms. Now, our focus will move to making the 'things' smarter using technologies like computer vision and deep learning on edge. This way, the things will be able to make informed decisions about their surroundings. That's IoT embedded vision.

Vision processing units (VPUs).

A VPU is a specific type of AI accelerator designed for computer vision tasks. It is well suited for parallel processing. A very popular edge AI accelerator is the Intel Neural Compute Stick 2, which is built on the Myriad X VPU. It offers an easy-to-use USB interface.

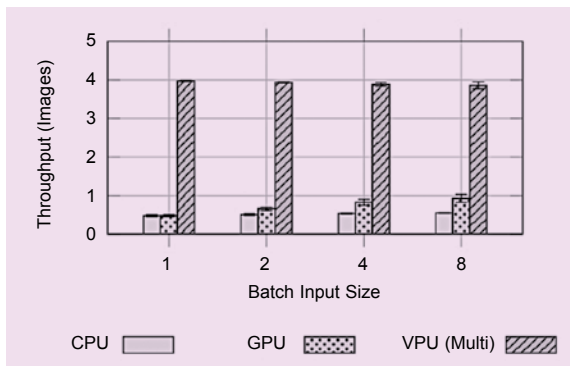
Pre-trained convolutional network (CNN) models are

used in the VPU chips to execute inference tasks. Studies have shown that VPUs are superior to CPUs and GPUs in terms of performance as well as power consumption. Moreover, a combination of multiple VPU chips can even reduce the thermal-design power (TDP) by up to eight times.

Software 2.0 and computer vision. As we move towards Industry 4.0, software development is also moving towards Software 2.0. Software 1.0—the technology stack that includes languages like C/C++, python, etc—is the one that we are familiar with. It needs a coder. On the other hand, Software 2.0 literally writes itself when we feed enough data to a neural network that knows how to code.

This is yet another reason why the computer vision market will continue to grow exponentially. It will drive Software 2.0. Andrej Karpathy, Director of AI at Tesla, is a big proponent of Software 2.0. He notes that with Software 2.0, the computer vision library (for example, OpenCV) could even be auto-tuned on our specific data! **EFY**

The author, Aaryaa Padhyegurjar, is an Industry 4.0 enthusiast at EFY with a keen interest in innovation and research.



Throughput performance comparison per watt using the CPU, GPU, and multi-VPU configurations (Credit: <https://arxiv.org>)